

1. Project Overview

This project focuses on building a Customer Ticket Analytics Dashboard to improve visibility into SLA performance, ticket resolution efficiency, and customer satisfaction for a customer support function. The solution combines Excel, SQL, and Tableau to transform raw ticket data into actionable insights and an interactive dashboard for managers and team leads.

2. Business Problem

Customer support leadership lacked a centralized, data-driven view of operational performance across teams and channels, leading to:

- Limited visibility into SLA adherence and breaches over time.
- Difficulty identifying low-performing teams and channels.
- No clear understanding of how resolution times affected SLA and CSAT outcomes.
- Inconsistent, anecdotal decision making instead of data-backed interventions.

These issues resulted in frequent SLA breaches, workload imbalance, and missed improvement opportunities.

3. Project Objectives

The project aimed to:

- Monitor SLA performance over time at weekly and team level.
 - Analyze team and channel efficiency in handling tickets.
 - Track customer satisfaction (CSAT) and correlate it with operations.
 - Identify root causes of SLA breaches (teams, channels, resolution delays).
 - Deliver an interactive Tableau dashboard tailored to stakeholders' decision needs.
-

4. Data Description

The dataset contained historical customer support ticket data with one row per ticket and the following key fields:

- TicketID, CreatedDate, WeekNumber
- Status (Open, Closed, Pending)
- Group (Billing, Tech, Tier1, Tier2, Escalations)
- Channel (Email, Phone, Portal)
- CSAT_Score, CSAT_Category (Positive, Neutral, Negative)
- ResponseTimeHrs, ResolutionTimeHrs
- SLA_Met (Yes/No)

The data covered multiple weeks of support operations, enabling both point-in-time metrics and trends.

5. Tools & Technologies

- Excel + Power Query: Initial data profiling, cleaning, and transformation.
 - SQL (SQLite): Aggregations and root-cause analysis for SLA and CSAT metrics.
 - Tableau: Design and implementation of the interactive dashboard and KPI views.
-

6. Data Preparation (Excel + Power Query)

Using Excel and Power Query, the following steps were performed:

- Handled missing values, standardized date formats, and normalized categories for Group, Channel, and Status.
- Created derived fields:
 - SLA_Flag (1 for SLA met, 0 for breached) from SLA_Met.
 - Resolution_Bucket (e.g., <24 hrs, 24–48 hrs, >48 hrs) based on ResolutionTimeHrs.
- Ensured correct data types (dates, numeric fields, text dimensions) and removed duplicate tickets.

The cleaned dataset was exported as a structured table suitable for SQL and Tableau.

7. SQL Analysis and Root Cause Exploration

SQLite was used to generate summarized metrics and uncover drivers of SLA performance.

7.1 SLA Performance

- Overall SLA percentage and weekly SLA trends were calculated.
- Example query to evaluate team performance:

```
SELECT
  "Group",
  ROUND(
    COUNT(CASE WHEN SLA_Met = 'Yes' THEN 1 END) * 100.0
    / COUNT(*),
    2
  ) AS SLA_Percentage
FROM tickets
GROUP BY "Group"
ORDER BY SLA_Percentage ASC;
```

This helped identify teams with the lowest SLA compliance and prioritize them for process review.

7.2 Channel and Operational Efficiency

- SLA compliance by Channel (Email, Phone, Portal) highlighted underperforming channels.
- Distribution of tickets by Status (Open, Closed, Pending) exposed backlog and work-in-progress load.
- Relationship between ResolutionTimeHrs, Resolution_Bucket, and SLA_Flag quantified the impact of delays on SLA breaches.

Insights from SQL guided the design of the dashboard views and filters.

8. Dashboard Design (Tableau)

The dashboard was designed for support managers and team leads, focusing on clarity, drill-down, and decision support.

8.1 Main Components

- KPI Cards: Overall SLA %, average CSAT, average Resolution Time.
- Trend View: Weekly SLA performance line chart to highlight dips and improvements.
- Operational Views:
 - Tickets by Group (team workload and SLA performance).
 - Ticket Status breakdown (Closed, Pending, Open) to monitor backlog.
- Customer Insights: CSAT distribution (Positive, Neutral, Negative) by Group and Channel.

8.2 Interactivity

- Filters for Week, Group, and Channel to enable focused analysis.
- Hover tooltips with SLA %, average resolution time, and CSAT for the selected slice.

The visual design followed best practices around minimal clutter, consistent color coding for status, and clear separation of KPIs, trends, and diagnostics.

9. Key Findings

Based on the analysis and Tableau visualizations:

1. SLA Performance
 - Overall SLA stood around 45–50%, meaning less than half of tickets met defined SLAs.
 - This indicates significant operational inefficiencies and capacity or process issues.
2. Trend Volatility
 - Weekly SLA performance fluctuated, with noticeable dips in specific weeks.

- These dips often correlated with higher ticket volumes or spikes in certain channels.
 - 3. Team Performance
 - Tier1 and Tier2 handled the majority of incoming tickets.
 - Some teams showed consistently lower SLA compliance, pointing to workload imbalance and possible process or skill gaps.
 - 4. Customer Satisfaction
 - CSAT was approximately 74% Positive and 26% Neutral/Negative.
 - While overall satisfaction was acceptable, the Neutral/Negative share suggested room for improvement, especially where SLA breaches were frequent.
 - 5. Resolution Time vs SLA
 - Tickets with ResolutionTimeHrs in higher buckets had disproportionately low SLA_Flag values.
 - This validated a strong link between delayed resolution and SLA failure, reinforcing the need to focus on cycle time.
-

10. Recommendations

Translating findings into actions:

- Improve SLA Compliance
 - Optimize ticket routing rules to balance load across Tier1, Tier2, and Escalations.
 - Introduce priority-based handling and clear SLAs per priority level.
- Balance Workload and Capacity
 - Redistribute incoming volume across teams based on historical SLA and capacity.
 - Invest in Tier1 automation and knowledge base to deflect simple queries.
- Enhance Channel Efficiency
 - Improve response handling for Email, which typically shows slower reaction and resolution times.
 - Explore chatbots and self-service portals to reduce manual load and accelerate first response.
- Reduce Resolution Time
 - Set automated escalation triggers for tickets nearing SLA breach.
 - Monitor delayed tickets via dedicated dashboard views and daily stand-ups.

Each recommendation directly links to metrics visible in the dashboard, ensuring traceability from insight to action.

11. Outcomes

The project delivered the following outcomes:

- A centralized Tableau dashboard for real-time monitoring of SLA, CSAT, and workload across teams and channels.
 - Improved transparency for stakeholders, enabling faster and more objective decision making.
 - Clear identification of low-performing teams, channels, and time periods, supporting targeted interventions.
 - A reusable data model and reporting framework that can be extended to additional KPIs (e.g., first response SLA, reopen rate).
-

12. Key Learnings and Reflection

From this project, the main learnings were:

- Thorough data cleaning and standardization in Excel/Power Query significantly reduces downstream complexity in SQL and Tableau.
- SQL is crucial to move from descriptive “what is happening” to explanatory “why it is happening” insights for SLA and CSAT.
- Effective dashboard design focuses on decision-making: surfacing the right KPIs and trends rather than adding excessive visuals.
- Interactivity (filters, tooltips, drill-downs) is essential for stakeholders to explore root causes independently and adopt the dashboard in their routine.